

Mechanical Design Engineer

Vaidya Precision Industries | Coimbatore, India | Full-time

About the role

We are seeking a Mechanical Design Engineer to lead the design of precision sheet metal and machined assemblies for our industrial automation product line. You will own designs from concept through prototyping and production release.

Key Responsibilities

- Produce 3D models and detailed 2D manufacturing drawings in SolidWorks
- Apply GD&T to drawings and run tolerance stack-up analysis on assemblies
- Perform finite element analysis to validate structural and thermal designs
- Select materials and specify heat treatment and surface finishing processes
- Work with the manufacturing team on CNC machining and sheet metal processes
- Create and maintain the bill of materials for each assembly
- Support prototyping, design validation testing and root cause investigations
- Drive design for manufacturability reviews with production and quality teams

Required Qualifications

- Bachelor's degree in Mechanical Engineering
- Minimum 4 years of experience in mechanical design for manufacturing
- Expert level SolidWorks and AutoCAD skills
- Strong working knowledge of GD&T and tolerance analysis
- Hands-on experience with finite element analysis using ANSYS
- Practical knowledge of CNC machining, sheet metal fabrication and welding
- Understanding of thermodynamics and heat transfer fundamentals
- Experience creating and releasing engineering drawings to production

Preferred Qualifications

- Experience with CATIA or Creo
- Six Sigma or lean manufacturing certification
- Exposure to computational fluid dynamics
- Experience in the automotive or aerospace sector
- Knowledge of hydraulics and pneumatics systems

What we offer

- Competitive salary and annual performance bonus
- On-site canteen and transport